

Project 2: Image Matching, Camera Pose Estimation, and 3D Reconstruction

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October 1, 2025

Instructions

1. Deadline: 2025-11-02 11:59 pm
2. You need to write your report in $\text{\LaTeX}$. Word is acceptable, but the final submission must be a PDF. All text must be in English.
3. Your report should be submitted in PDF format and packed with your code. The naming format of the compressed file should be `studentID-name-project2.zip`.
4. Please submit your zip file to the designated course platform or link (e.g., <https://epan.shanghaitech.edu.cn/1/iFs6h>).

1 Introduction

This project is designed to provide hands-on experience with fundamental concepts in computer vision, specifically focusing on local feature matching, camera pose estimation, and 3D reconstruction. You will start by creating your own dataset, processing it with Structure from Motion (SfM) techniques using COLMAP, and then evaluating a state-of-the-art feature matching algorithm, SuperGlue. The advanced portion of this project will challenge you to compare multiple recent methods for image matching and 3D reconstruction across a variety of challenging real-world scenarios, including aerial and ground-level imagery.

2 Basic Tasks

2.1 Data Acquisition

Capture a sequence of 20 images of a scene of your choice using a smartphone or digital camera. Ensure that there is significant overlap between consecutive images to allow for robust feature matching. You can optionally select some challenging scenarios.

2.2 Structure from Motion with COLMAP

Use COLMAP (<https://colmap.github.io/>) to process your 20 images. The goal is to reconstruct the camera poses (both intrinsic and extrinsic parameters) for each image. You should follow the standard COLMAP workflow:

1. Feature extraction.
2. Feature matching.

3. Incremental reconstruction.

After a successful reconstruction, you will have a sparse 3D model of your scene and the corresponding camera poses.

2.3 Test Set Creation

Your task is to create a custom test set for evaluating a feature matching algorithm. This test set should be structured similarly to the official test datasets used for methods like SuperGlue.

1. From your 20 images, create 10 image pairs with varying degrees of overlap and viewpoint change.
2. For each image pair, use the camera poses obtained from COLMAP to calculate the relative pose (relative rotation and translation) between the two images.
3. Organize these image pairs, their corresponding intrinsic parameters, and the ground truth relative poses into a test file. You can refer to the data loading scripts in the SuperGlue repository for the exact format: <https://github.com/magic Leap/SuperGluePretrainedNetwork>.

2.4 Evaluation with SuperGlue

Use the pre-trained SuperGlue model to perform inference on your custom test set. The SuperGlue paper[3] and its accompanying code will be your primary resources.

1. Run SuperGlue on your 10 image pairs to find correspondences.
2. Evaluate the performance of SuperGlue. This should include quantitative metrics, such as the pose error calculated from the matches found by SuperGlue versus your ground truth poses from COLMAP.
3. Provide a qualitative analysis. Include visualizations of the matches found by SuperGlue on your image pairs.
4. Discuss the results. Analyze the successes and failures of SuperGlue on your dataset. In which cases did it perform well, and in which cases did it fail? What are the potential reasons for these outcomes (e.g., large viewpoint changes, low texture, repetitive patterns)?

3 Advanced Tasks

For the advanced tasks, you will conduct a comparative study of several state-of-the-art methods for image matching and 3D reconstruction across different challenging scenarios.

3.1 Comparative Analysis of Matching Methods

Compare the performance of the following three image matching methods:

1. **Superpoint+Superglue:** <https://github.com/magic Leap/SuperGluePretrainedNetwork>
2. **MASt3R:** <https://github.com/naver/mast3r> [2]
3. **VGGT:** <https://github.com/facebookresearch/vgggt> [4]

Your comparison should be performed across the following three scenarios:

- **Scenario 1: Drone Nadir View.** Use consecutive frames from a drone flying with a vertical (nadir) view. You can source this data from datasets like MatrixCity [5].
- **Scenario 2: Drone Oblique View.** Use consecutive frames from a drone flying with an oblique (tilted) view. This data can be found in datasets like Horizon_GS [1], or you can create your own by splitting a drone video from the web or taking pictures from a high vantage point.

- **Scenario 3: Air-to-Ground View.** Match images of the same scene captured from a drone’s perspective and a ground vehicle’s perspective. You can find such pairs in datasets like MatrixCity, UC-GS, or Horizon.GS, or capture your own.

For each scenario, present a qualitative and quantitative comparison of the three methods. Discuss their strengths and weaknesses in each context.

3.2 Comparative Analysis of Localization and Reconstruction

Compare the performance of MAST3R and VGGT for image localization and point cloud reconstruction. Your analysis should cover the same three scenarios as in the previous section. For Scenario 3 (Air-to-Ground), you should test with both a single air-to-ground image pair and with multiple consecutive air-to-ground pairs.

3.3 Report and Analysis

In your final report, you must include:

- Visualizations of the image matching results from all methods and scenarios.
- Visualizations of the point cloud reconstruction results.
- A thorough performance analysis.

4 Submission

You should submit a single PDF report containing all your results, visualizations, and analysis. The report should be well-structured and clearly written. Please also include a link to your code, which should be organized and commented.

References

- [1] Lihan Jiang et al. “Horizon-GS: Unified 3D Gaussian Splatting for Large-Scale Aerial-to-Ground Scenes”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2025, pp. 26789–26799.
- [2] Vincent Leroy, Yohann Cabon, and Jerome Revaud. *Grounding Image Matching in 3D with MAST3R*. 2024. arXiv: 2406.09756 [cs.CV].
- [3] Paul-Edouard Sarlin et al. “SuperGlue: Learning Feature Matching with Graph Neural Networks”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2020. URL: <https://arxiv.org/abs/1911.11763>.
- [4] Jianyuan Wang et al. “VGGT: Visual Geometry Grounded Transformer”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2025.
- [5] Saining Zhang et al. “Drone-assisted Road Gaussian Splatting with Cross-view Uncertainty”. In: *arXiv preprint arXiv:2408.15242* (2024).